INCH-POUND

MIL-V-5027D
AMENDMENT 2
10 January 1996
SUPERSEDING
AMENDMENT 1
1 February 1980

MILITARY SPECIFICATION

VALVES CHECK, OXYGEN, HIGH PRESSURE

This amendment forms a part of MIL-V-5027D, dated 12 March 1968, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Add the following to the bottom of page 1:

"Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Systems Engineering Department, Code 4.1.4B120-3, Naval Air Warfare Center Aircraft Division, Highway 547, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter."

PAGE 2

2.1 Specifications - Federal and Military

Delete the following:

"O-T-235 Tetrachloroethelene (Perchloroethylene), Technical Grade
O-T-634 Trichloroethylene, Technical
* MIL-C-81302 Cleaning Compound, Solvent, Trichlorotrifluoroethane
* MIL-T-81533 Trichloroethane 1, 1, 1 (Methyl Chloroform) Inhibited, Vapor Degreasing"

PAGE 3

2.1 Standards - Military

Delete "MS33586 Metals, Definition of Dissimilar"

Add the following:

"MIL-STD-889 Dissimilar Metals
* MIL-STD-1359 Cleaning Methods and Procedures for Breathing Oxygen Equipment"

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PAGE 3 (Continued)

Standards - Federal

Add "FED-STD-H28 Screw-Thread Standards for Federal Services."

2.2 Other Publications

Delete "National Bureau of Standards Publication Handbook H28 Screw-Thread Standards for Federal Services.
(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)"

PAGE 4

3.1 Delete and substitute the following:

"3.1 Qualification - The valves furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3, 4.3.1, and 6.4)."

3.2 Delete and substitute the following:

"3.2 First article - When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4, 4.4.1 and 6.2(c))."

3.3.1.1, line 3: Delete "MS33586" and substitute "MIL-STD-889."

PAGE 5

3.4.1.3 Delete "Handbook H28" and substitute "FED-STD-H28."

* 3.4.1.5 Delete and substitute the following:

"3.4.1.5 Degreasing - Prior to assembling the valve, all internal surfaces and components shall be cleaned in accordance with MIL-STD-1359. After assembly, the absence of cleaning compounds shall be verified in accordance with MIL-STD-1359."

PAGE 6

* 3.5.12 Delete "3.5.12" and substitute "3.5.12.1."

* Add new paragraph "3.5.12 Oxygen test."

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PAGE 6 (Continued)

* Add new paragraph "3.5.12.2 Oxygen purge - When tested in accordance with 4.7.13.2, the valve shall not ignite, and there shall be no evidence of charring or deterioration."

PAGE 7

4.1 Delete and substitute the following:

"4.1 Responsibility for inspection - Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

PAGE 8

4.2(a) Delete and substitute the following:

"a. Qualification inspection - Qualification inspection consists of examinations and tests performed on samples submitted for approval as qualified products (see 3.1, 4.3, 4.3.1, and 6.4).

(1) Retention of qualification - Retention of qualification consists of periodic verification to determine compliance of the qualified product with the requirements of this specification (see 3.1, 4.3, and 4.3.2)."

4.3.1, lines 5 and 6: Delete "the Supply Officer, Naval Air Development Center, Johnsville, Warminster, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department." and substitute "a test facility set forth in the letter of authorization to submit samples (see 6.4)."

4.3.2 Add the following new paragraph:

"4.3.2 Retention - The retention of qualification shall consist of periodic verification to determine compliance of the qualified valves with the requirements of this specification. The time and method of periodic verification shall be specified by the activity responsible for the Qualified Products List and shall be included in the Notice of Qualification letter."

* 4.4 Add to bottom of list "Oxygen purge, if required."

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PAGE 9

4.4.1, 4.4.1.1, 4.4.1.1.1 and 4.4.1.2 Delete and substitute the following:

"4.4.1 First article samples - Unless otherwise specified as soon as practicable after award of a contract or order the manufacturer shall submit four valves of each MS or AN number specified in the contract or order. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these valves from contract to contract, submission of further first article samples on the new contract may be waived at the direction of the procuring activity (see 6.2). Approval of the first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article samples shall be furnished to the government as directed by the contracting officer (see 6.2).

4.4.1.1 Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the government quality assurance representative. Two approved valves will be returned to the manufacturer for use in monitoring production. The other valves will be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract."

* 4.5 Add to bottom of list "Oxygen purge, if required."

PAGE 11

* Add to the bottom of table II:

"

Oxygen purge	4.7.13.2	Every valve	Reject all defective units

"

* 4.6.1 After first sentence, add: "If oxygen is not used as the test gas, oxygen in accordance with MIL-O-27210, type I shall be used to purge in accordance with 3.5.12.2 and 4.7.13.2."

PAGE 15

* 4.7.13 Delete "4.7.13" and substitute "4.7.13.1."

* Add new paragraph "4.7.13 Oxygen test."

* 4.7.13.1 Delete "4.7.13.1" and substitute "4.7.13.1.1."

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* 4.7.13.2 Delete "4.7.13.2" and substitute "4.7.13.1.2."

* 4.7.13.3 Delete "4.7.13.3" and substitute "4.7.13.1.3."

* Add new paragraph "4.7.13.2 Oxygen purge - The valve shall be connected to a regulated source of oxygen (see 4.6.1). An inlet pressure of 1800 psig shall be applied and the valve shall free flow for 3 to 5 minutes. After testing, the valve shall meet the requirements of 3.5.12.2."

5. Delete "PREPARATION FOR DELIVERY" and substitute "PACKAGING."

PAGE 21

6.3 Delete paragraph in its entirety and substitute the following:

"6.3 Contract data requirements - When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-1049(n) of the Defense Acquisition Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements list (DD Form 1423) incorporated into the contract when the provisions of DAR-7-1049(n) are not involved, the data specified below will be delivered by the contract in accordance with the contract requirements deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
4.4.1.1	First Article Inspection Reports	DI-T-5329 Inspection Test Reports"

6.4, line 12: Delete "Johnsville."

lines 12 and 13: Delete "Director, Aerospace Crew Equipment Department" and substitute "Code 6031."

6.5 Add the following new paragraph:

"6.5 When a first article is required, it shall be tested under the appropriate provision of 7-10455 of the Defense Acquisition Regulation. The first article should be a preproduction sample. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test, and approval of the first article."

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